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Peer-Review Worksheet Pack

ORBIT NORTH · Week 3 observer worksheets — four axes

ORBIT NORTH
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Faculty of Engineering · School of Engineering Design and Teaching Innovation (SEDTI)

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Classification: Internal

Peer-Review Worksheet Pack

You are the observer team. While the operator team flies the shift, **you build the evidence** the cohort uses at the post-sim debrief. Each worksheet below is a single page, used once per observed shift. Use the worksheet for the axis the instructor assigns you; do not try to fill in all four for the same shift.

The four axes:

1. **Team** — role clarity, FD authority, coverage, fatigue.
2. **Communications** — voice loop, callouts, handover, MCS chat.
3. **Space usage** — console layout, line of sight, distractions, physical comfort.
4. **Technical** — procedure adherence, anomaly response timing, telemetry interpretation, command discipline.

How to use a worksheet

Each worksheet has four blocks:

Block	What to write
Observed	Facts. Timestamps, callouts, telemetry values, exact words. **No judgement here.**
Worked well	One or two specific behaviours that improved the shift.
Could improve	One or two specific behaviours that hurt the shift.
Specific recommendation	What you'd change *concretely* for next time. One sentence, actionable.

Rules:

- Be specific. "Team didn't communicate" fails. "At T+34 the AOCS position commanded a wheel spin without an FD GO" passes.
- Use timestamps. The simulator clock is on every screen.
- Complete the worksheet **after** the shift. Do not show your draft to the operator team during the shift.
- The instructor reads worksheets at the debrief. Be direct. The goal is to make the operator team better, not to be polite.

Worksheet 1 — Team

Observer: __ Shift / pass: _ *Date:* ___ Operator team: _____

Observed

Write one note per row: a timestamp from the simulator clock, what happened, and who was involved.

Item	Note (timestamp · what · who)
Did each position know its role?	
Was the FD authoritative?	

Item	Note (timestamp · what · who)
Was anyone over-loaded or under-utilised?	
Did anyone do another position's job?	
Was fatigue handled (breaks, water)?	
How did the team handle disagreement?	

Worked well

(one or two specific behaviours)

Could improve

(one or two specific behaviours)

Specific recommendation

(one actionable sentence)

Worksheet 2 — Communications

Observer: __ Shift / pass: _____

Observed

Item	Note
Voice-loop discipline (callouts, hot-mike)	
GO/NO-GO protocol followed	
Cross-talk incidents (count + timestamps)	
MCS chat — used for what?	
Handover: timing, content, completeness	
Any silent confusion (operator did the wrong thing because no one said anything)?	

Worked well

(one or two specific behaviours)

Could improve

(one or two specific behaviours)

Specific recommendation

(one actionable sentence)

Worksheet 3 — Space usage

Observer: __ Shift / pass: _____

Observed

Item	Note
Console layout — could every position see what they needed?	
Line of sight — could the FD see all positions?	
Distractions (phones, side conversations, music)?	
Lighting, temperature, noise — issues?	
Movement around the room (procedure printouts, water)	
Physical comfort — chairs, monitors, posture	

Worked well

(one or two specific behaviours)

Could improve

(one or two specific behaviours)

Specific recommendation

(one actionable sentence)

Worksheet 4 — Technical

Observer: __ Shift / pass: _____

Observed

Item	Note
Procedure adherence — any skipped or out-of-order steps?	
Anomaly detection time — from first indicator to first action	
Telemetry interpretation — any misreads?	
Command discipline — any unauthorised commands?	
Use of the FDIR panel — did anyone watch it?	
Outcome vs pass criteria	

Worked well

(one or two specific behaviours)

Could improve

(one or two specific behaviours)

Specific recommendation

(one actionable sentence)

Worksheet rotation

Each pass, observer team uses **one** worksheet axis. Suggested rotation across the simulation week:

Day	Pass	Observer team uses
Wed (LEOP)	Pass 1	Communications
Wed (LEOP)	Pass 2	Technical
Thu (Commissioning)	Pass-set 1	Team
Thu (Commissioning)	Pass-set 2	Space usage

This way every axis gets fresh observation across the week. At the Friday wrap, each team holds two completed worksheets — those go into the cohort lessons-learned register.

After the shift

1. Take 10 minutes alone with the worksheet before the debrief starts. Re-read what you wrote; tidy it.
2. At the debrief, the FD speaks first. Then your team reads its worksheet aloud. The operator team responds with one *accept* and one *defend* per axis.
3. Hand the worksheet to the instructor at end of debrief.

What good observation looks like

> **TODO (author):** Write the 250-word worked example: a completed *Communications* worksheet for an LEOP pass, with specific, timestamped observations, one or two behaviours that worked and one or two that did not, ending with one concrete recommendation. It is the model students are given on Day 1.

References

- ECSS-E-ST-70-32C — Test and operations procedures — https://ecss.nl/standards/active-standards/ecss-e-st-70-32c-test-and-operations-procedures/
- *Crew Resource Management* (FAA Advisory Circular 120-51E), applicable to multi-position crew coordination — https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/22019
- NASA *Spaceflight Crew Operations* materials — https://appel.nasa.gov/